

Yiyang CAI

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🌐 YIYANGCAI | 📄 Google Scholar

Hong Kong University of Science and Technology - Clear Water Bay - Hong Kong

EDUCATION

- **Hong Kong University of Science and Technology** Sep. 2024 - Current
Ph.D. Student of Independent Interdisciplinary Program (Supervisor: Prof. Wenhan LUO and Prof. Yike GUO) Kowloon, HKSAR
◦ Research Interests: Image/Video Generation, Multimodal Learning, World Models
- **University of California, Berkeley** Aug. 2020 - Dec. 2021
Master of Engineering, Visual Computing and Computer Graphics, (Supervisor: Prof. Brian BARSKY) Berkeley, CA, United States
◦ GPA: 3.65/4.00
- **Beihang University** Aug. 2016 - Jun. 2020
Bachelor of Engineering, Automation (Shenyuan Honor School) Beijing, China
◦ GPA: 3.81/4.00 (Outstanding Graduate Honor)

PUBLICATIONS (FIRST OR CO-FIRST ARE LIGHTED IN RED)

- [1] **Y. Cai***, N. Chen*, R. Xie, J. Pan, C. Jiang, C. Chen, C. Liu, Z. Sun, W. Xue, W. Luo & Y. Guo. HOMIE: Human-object Centric Video Personalization via Multimodal Intelligent Enhancement. The 40th Annual Conference on Neural Information Processing Systems, 2026. (CCF-A)
- [2] N. Chen*, **Y. Cai***, R. Xie, J. Pan, C. Chen, W. Jia, Z. Cheng, C. Liu, Z. Sun & W. Luo. DomainShuttle: Freeform Open Domain Subject-driven Text-to-video Generation. Under Review, 2026.
- [3] C. Jiang, Y. Zhang, **Y. Cai**, C. Chan, Y. Liu, M. Chen, W. Xue & Y. Guo. Semantic Voting: A Self-Evaluation-Free Approach for Efficient LLM Self-Improvement on Unverifiable Open-ended Tasks. The 14th International Conference on Learning Representations, 2026. (CCF-A)
- [4] **Y. Cai**, Z. Jiang, Y. Liu, C. Jiang, W. Xue, Y. Guo & W. Luo. Foundation Cures Personalization: Improving Personalized Models' Prompt Consistency via Hidden Foundation Knowledge. The 39th Annual Conference on Neural Information Processing Systems, 2025. (CCF-A)
- [5] C. Jiang, C. Chan, **Y. Cai**, Y. Liu, W. Xue, & Y. Guo. Graceful Forgetting in Generative Language Models. Empirical Methods in Natural Language Processing (Main Conference), 2025. (CCF-B)
- [6] W. Cheng, W. Zhang, H. Shen, **Y. Cai**, X. He, K. Lv, & Y. Liu. Optimize weight rounding via signed gradient descent for the quantization of LLMs. Empirical Methods in Natural Language Processing, 2024.
- [7] H. Chang, H. Shen, **Y. Cai**, X. Ye, Z. Xu, W. Cheng, K. Lv, W. Zhang, Y. Lu & H. Guo. Effective quantization for diffusion models on CPUs. NeurIPS 2023 Workshop on Diffusion Models, 2023.
- [8] W. Cheng, **Y. Cai**, K. Lv, & H. Shen. TEQ: Trainable equivalent transformation for quantization of LLMs. arXiv preprint arXiv:2310.10944, 2023.
- [9] L. Tang*, **Y. Cai***, J. Liu, Z. Hong, M. Gong, M. Fan, J. Han, J. Liu, E. Ding, & J. Wang. Few-shot font generation by learning fine-grained local styles. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (pp. 7895-7904), 2022. (CCF-A)

PROJECTS

- **Multi-subject-driven Video Generation** Aug. 2025 - Aug. 2026
Keyword: Personalized Diffusion Models, Video Generation, Human-object Interaction 🔗
◦ Completed a comprehensive literature review of video generation approaches.
◦ Inspired by unified understanding and generation models, proposed a MLLM integration strategy to enhance video personalization models in more challenging scenarios, including multi-subject video personalization, abstract concept personalization.
◦ Conducted comprehensive comparison to popular video personalization models, validated the advantages of the proposed method and the corresponding paper submission is now accepted by NeurIPS 2026 as a poster.

• Training-free Prompt Following Optimization for Facial Personalization Models

Sep. 2024 - Apr. 2025

Keyword: Personalized Diffusion Models, Prompt Following, Training-free



- Completed a comprehensive literature review of facial personalization approaches.
- Identified the current limitation shared by state-of-the-art facial personalization methodologies: poor prompt following performance on fine-grained facial attributes (e.g., hair style, accessories, expression).
- Proposed a dual-denoising branch in personalization models (w./w.o. identity information) with an optimized self-attention modules that transfers satisfying facial attributes into personalization denoising process.
- Validated the proposed method on popular personalization models built upon different foundation models (e.g., SDv1.5, SDXL, FLUX.1-dev).
- Finished a paper submission and the paper has been accepted by NeurIPS 2026.

WORKING EXPERIENCE

• ByteDance Inc. (China)

Aug. 2025- Sep. 2026

Algorithm Intern

Beijing, China

- Explore video personalization under the enhancement of MLLMs.
- Explore infinite video generation under auto-regressive architecture and its application in video personalization, including interactive reference insertion.

• Intel Inc. (China)

Jun. 2022- Jun. 2024

Machine Learning Engineer, Group of Data Center and Artificial Intelligence

Shanghai, China

- Developed and maintained key features of [Intel Neural Compressor](#), including quantization and pruning for diffusion models and large language models.
- Developed efficient GPTQ API for a large variety of LLMs and VLMs (e.g., MPTs, Qwens, ChatGLMs, Baichuan, LLaVA). Designed pruning API to support different orthogonal configurations (structures, criterion, schedulers, etc.). Validated API's function on a wide range of model, including CV, NLP models and popular LLMs.

• Baidu Inc.

Sep. 2020 - Nov. 2021

Algorithm Intern, VIS Group

Beijing, China

- GAN-based style transfer research for few-shot Chinese font generation: given a small set of style-specified glyphs (≤ 100 examples), generated the entire personalized font ($\approx 20K$ instances). Applied a novel attention mechanism to improve the model's ability of capturing fine-grained style and spatial feature representations from glyphs. Designed character decomposition strategy to boost efficiency of learning characters' component representations.

SKILLS

- **Programming Languages:** Python, C++
- **Deep Learning & Computer Vision Tools:** PyTorch, DeepSpeed, Matplotlib, OpenCV and Pandas
- **Language Proficiency:** TOEFL: 104 (Reading 28 + Listening 25 + Speaking 26 + Writing 25)

TEACHING EXPERIENCE

• Teaching Assistant

Feb. 2026 - May. 2026

TA duty for EMIA-4110 (Practical Machine Learning) (Teacher: Prof. Wenhan LUO)

HKUST

- Delivered several presentations of advanced artificial intelligence.
- Supported the professor's course preparation of image generation tutorials.

• Teaching Assistant

Sep. 2025 - Dec. 2025

TA duty for MAIE-5421 (Computer Vision) (Teacher: Prof. Wenhan LUO)

HKUST

- Delivered several presentations of advanced artificial intelligence.
- Supported the professor's course preparation of tutorials, project designing and final grading.

• Teaching Assistant

Feb. 2025 - May. 2025

TA duty for EMIA-4110 (Practical Machine Learning) (Teacher: Prof. Wenhan LUO)

HKUST

- Delivered several presentations of using Colab and PyTorch to undergraduate students.
- Supported the professor's course preparation of image generation tutorials.